

BIOGRAPHICAL SKETCH

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NAME: Vance, Michaela

eRA COMMONS USER NAME (credential, e.g., agency login): MVANCE2

POSITION TITLE: Graduate Student Research Assistant (PhD Candidate)

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	Start Date MM/YYYY	Completion Date MM/YYYY	FIELD OF STUDY
Rose State College Midwest City, OK	Not Applicable	08/2017	05/2018	Dual Enrollment
University of Central Oklahoma Edmond, OK	Bachelors, BS	08/2018	05/2022	Biology-Biomedical Sciences
University of Oklahoma Health Sciences Center Oklahoma City, OK	PhD	07/2022	12/2027	Cell Biology

A. Personal Statement

My interest in science began during my senior year of high school while performing a titration in my AP Chemistry class. Growing up in a small town in Oklahoma, I had never been made aware of opportunities involving scientific research until my chemistry teacher, who had previously worked at a research foundation, introduced me to the idea. I had always had a passion for learning new things, asking questions, and problem-solving, so the idea that these qualities could lead to a career in biomedical research was exciting to me.

After my freshman year of college, I began working full-time in Dr. Nikki Seagraves's lab at the University of Central Oklahoma. We investigated fetal cardiovascular malformations resulting from maternal phenylketonuria (MPKU). During my time in the lab, I worked on projects to determine if phenylalanine acts as an inhibitor of the retinoic acid signaling pathway in neural crest cells which form the developing heart. During my three years with Dr. Seagraves, I learned mammalian cell culture, microscopy, DNA and RNA isolation, PCR, gel electrophoresis, and cloning. Additionally, I worked independently and developed skills involving interpreting scientific literature, developing experimental protocols, analyzing data, and presenting scientific findings. This experience fostered my aspirations in biomedical research and piqued my interest in vascular biology.

In 2022, I joined the Graduate Program in Biomedical Sciences at the University of Oklahoma Health Sciences Center and Dr. Shannon Conley's lab. We study mechanisms of cerebrovascular aging, and my work on various projects in Dr. Conley's lab has led to me being a contributing author on four primary research articles. I was interested in joining Dr. Conley's lab because it allowed me to pivot from the study of vascular development to vascular aging, providing me with a unique perspective in the field of vascular biology.

For my dissertation work, I am investigating the mechanisms underlying how age-related increases in brain senescence drive endothelial to mesenchymal transition (EndMT) in the brain microvasculature leading to cerebrovascular fragility and dysfunction. Components of the senescence-associated secretory phenotype can trigger EndMT. During EndMT, endothelial cells lose characteristic endothelial phenotypes, including some that are critical for cerebrovascular function, and transition to a mesenchymal-like phenotype. However, the role of EndMT in brain aging, and its contributions to cerebrovascular fragility, are unexplored. During my time in Dr. Conley's lab, I have developed new techniques that will be crucial for my dissertation work such as immunofluorescent labeling and microscopy, mouse surgical techniques and tissue collection, RT-qPCR, mouse neurological assessment, and bioinformatic analysis. Additionally, I have continued to sharpen my scientific communication skills by presenting my research and working on scientific writing.

My previous research experience and steadfast dedication to advancement in the field of vascular aging have made me a well-suited candidate for a future career in vascular aging research. After my dissertation work, I will apply for, and complete, a postdoctoral fellowship to help me transition into a skilled, independent scientist. My long-term goal is to study how age-related increases in inflammation and oxidative stress can result in the reprogramming of endothelial cells contributing to the instability of cerebral microvessels.

B. Positions, Scientific Appointments and Honors

Positions and Scientific Appointments

2025-present	President, Graduate Student Association, University of Oklahoma Health Sciences Center
2024-present	Vice President, Graduate Student Association, University of Oklahoma Health Sciences Center
2024-present	Member, Crimson Club, University of Oklahoma Health Sciences Center
2024-present	Member, Academic Appeals Board, University of Oklahoma Health Sciences Center
2023-present	Member, American Aging Association
2022-present	Graduate Research Assistant, University of Oklahoma Health Sciences Center
2023-2024	Senator, Graduate Student Association, University of Oklahoma Health Sciences Center
2023-2024	Senator, Student Government Association, University of Oklahoma Health Sciences Center
2022-2023	Member, Graduate Student Association, University of Oklahoma Health Sciences Center
2021-2022	Cell Biology Supplemental Instructor, University of Central Oklahoma
2021-2022	President, Beta Beta Beta Biological Honor Society, University of Central Oklahoma
2020-2021	Vice President, Beta Beta Beta Biological Honor Society, University of Central Oklahoma
2020-2022	Member, Society for Developmental Biology
2019-2022	Undergraduate Research Assistant, University of Central Oklahoma
2019-2020	Sergeant of Arms, Beta Beta Beta Biological Honor Society, University of Central Oklahoma

Honors

2025	Trainee Data Blitz Award, American Aging Association 53 rd Annual Meeting
2025	College of Medicine Dean's Award for Scientific Achievement (Travel Award), GREAT Symposium, University of Oklahoma Health Sciences Center
2024	Best Graduate Student Poster Presentation (Travel Award), Cell Biology Symposium, University of Oklahoma Health Sciences Center
2022	Summa Cum Laude, University of Central Oklahoma
2022	Outstanding Senior Award, Department of Biology, University of Central Oklahoma
2022	Academic Achievement Award, Department of Biology, University of Central Oklahoma
2021	Dr. Ethel Derrick Endowed Scholarship for Biology, University of Central Oklahoma
2021	Selected for a Posters on the Hill Poster Presentation, Council on Undergraduate Research
2021	Best Student Poster Finalist (Top 6), Society for Developmental Biology 80 th Annual Meeting
2021	Summer Undergraduate Research Program Participant, University of Oklahoma Health Sciences Center
2021	Frank G. Brooks Award for Outstanding Oral Presentation (2 nd Place), Beta Beta Beta Biological Honor Society South-Central Regional Convention
2020-2022	Awarded Research, Creative, and Scholarly Activities Grants, University of Central Oklahoma
2018-2022	President's Honor Roll, University of Central Oklahoma

C. Contributions to Science

Undergraduate Research:

1. The Effect of Phenylalanine on the Retinoic Acid Signaling Pathway During MPKU:

Phenylketonuria (PKU) is a genetic disorder that results in phenylalanine hydroxylase reduction. When a pregnant person with PKU does not adhere to a phenylalanine restricted diet during pregnancy, elevated levels of phenylalanine can cross the placenta acting as a teratogen to the fetus resulting in maternal PKU (MPKU). Cardiovascular malformations are common in MPKU; however, the mechanisms underlying how high phenylalanine concentrations result in malformations are not well understood. Transcriptomic analysis revealed alterations in retinoic acid metabolism during critical stages of chick heart development after treatment with phenylalanine. Thus, we hypothesized that phenylalanine acts as a retinoic acid pathway inhibitor resulting in

decreased proliferation of neural crest cells which help form the outflow tract and aortic arch arteries of the heart. To test this, I treated mouse O9-1 Cranial Neural Crest Cells with phenylalanine, retinoic acid, retinal, and citral (a retinoic acid inhibitor) and measured proliferation. We found that phenylalanine decreased proliferation like citral suggesting that it may act as a retinoic acid pathway inhibitor. As part of related work, I also cloned the ADH1C enzyme for use in future enzyme kinetic assays to determine the effect of phenylalanine on metabolism of retinol by alcohol dehydrogenase. I was responsible for all cell culture, treatment, labeling, fluorescent microscopy, and data analysis during these experiments. Findings from these studies further the understanding of the mechanisms underlying phenylalanine teratogenicity and cardiovascular malformations in MPKU.

- a. **Vance ML***, Nurbyek N*, Smith G, Seagraves NJ. Effect of phenylalanine, retinoic acid, retinal, and citral on the proliferation of O9-1 mouse cranial neural crest cells. Council on Undergraduate Research's Posters on the Hill; 2021 April; Virtual.
- b. Nurbyek N*, **Vance ML***, Seagraves NJ. Effect of phenylalanine, retinoic acid, retinal, and citral on the proliferation of O9-1 mouse cranial neural crest cells. Society for Developmental Biology 80th Annual Meeting; 2021 July; Virtual.
- c. **Vance ML**, Seagraves NJ. Cloning of Alcohol dehydrogenase 1C for enzyme kinetics. Society for Developmental Biology 81st Annual Meeting; 2022 July; Vancouver, Canada.

Graduate Research:

2. The Effects of Time Restricted Feeding on the Aged Mouse Retina:

Time restricted feeding (TRF) is a dietary modification in which feeding is restricted to a portion of the day without changes in caloric intake. TRF has vasoprotective effects in the aged brain leading to improvements in age-related cognitive performance; however, no studies have examined the effects of TRF on the aged retina. We hypothesized that mice on a TRF diet would have improvements in age-related declines in retinal function and increases in inflammation. Changes in the retina were examined using electroretinography, fluorescein angiography, histology, and bulk RNA sequencing. I contributed to this project by preparing retinal flat mounts and labeling them markers of vasculature and microglia to be used for analysis of vascular density, vascular network characteristics, and microglial activation. Additionally, I helped to gather and analyze bulk RNAseq data and participated in writing. We found that TRF does not affect age-related signs of retinal inflammation or microglial activation but does help to preserve retinal function in aged mice.

- a. Huston CA, Milan M, **Vance ML**, Bickel MA, Miller LR, Negri S, Hibbs C, Vaden H, Hayes L, Csiszar A, Ungvari Z, Yabluchanskiy A, Tarantini S, Conley SM. The effects of time restricted feeding on age-related changes in the mouse retina. *Experimental Gerontology*. 2024 Jul. 5;194:112510. Epub ahead of print. PMID: PMC11425985.

3. Mechanisms of IGF-1 Signaling on Vascular Smooth Muscle Cells and its Contribution to Cerebrovascular Dysfunction:

Insulin-like growth factor 1 (IGF-1) is a pleiotropic growth hormone that has a vasoprotective role. IGF-1 levels decline with age, and mouse models of circulating IGF-1 deficiency mimic phenotypes seen in cerebrovascular aging. Our lab is interested in understanding the contributions of altered IGF-1 signaling on vascular smooth muscle cells (VSMCs) to the development of age-related vascular pathologies (such as cerebral microhemorrhage formation, impaired autoregulation, impaired neurovascular coupling, etc.) and associated cognitive decline. We hypothesize that age-related alterations in IGF-1 lead to VSMC dysfunction resulting in vascular fragility contributing to vascular cognitive impairment and dementia (VCID). To test this hypothesis, we have utilized VSMC cell cultures treated with IGF-1 and a VSMC-specific IGF-1 receptor deficient mouse model. I contributed to this project by performing expression analysis (microfluidic qRT-PCR) for genes associated with microhemorrhages, neurovascular coupling, and angiogenesis. I have helped analyze gene expression data, microbleed data, and gait data, performed neurological assessments, and participated in scientific writing. Overall, we found that decreased IGF-1 signaling on VSMCs contributes to cerebrovascular fragility.

- a. Bickel MA, Sherry DM, Bullen EC, **Vance ML**, Jones KL, Howard EW, Conley SM. Microvascular smooth muscle cells exhibit divergent phenotypic switching responses to platelet-derived growth factor and insulin-like growth factor 1. *Microvascular Res*. 2024 Jan. 151:104609. PMID: PMC10842624.
- b. Miller LR*, Bickel MA*, Tarantini S, Runion ME, Matakchiera Z, **Vance ML**, Hibbs C, Vaden H, Nagykalai D, Martin T, Bullen EC, Pinckard J, Kiss T, Howard EW, Yabluchanskiy A, Conley SM. IGF1R deficiency in vascular smooth muscle cells impairs myogenic autoregulation and cognition in mice. *Frontiers in Aging Neuroscience*. 2024 Feb 15. 16:1320808. PMID: PMC10902040.

- c. Miller LR, Bickel MA, **Vance ML**, Vaden H, Nagykalai D, Nyul-Toth A, Bullen EC, Gautam T, Tarantini S, Yabluchanskiy A, Kiss T, Ungvari Z, Conley SM. Vascular smooth muscle cell-specific Igf1r deficiency exacerbates the development of hypertension-induced cerebral microhemorrhages and gait defects. *Geroscience*. 2024 Jun. 46(3):3481-3501. PMID: PMC11009188.
- d. **Vance ML**, Miller LR, Tarantini S, Yabluchanskiy A, Ungvari Z, Csiszar A, Conley SM. The role of IGF-1 on vascular smooth muscle cells in the development of cerebral microhemorrhages. American Aging Association 51st Annual Meeting; 2024 June; Oklahoma City, Oklahoma.

4. Senescence and Endothelial to Mesenchymal Transition in the Aging Cerebral Vasculature:

Senescent cells increase with age and acquire a senescence-associated secretory phenotype (SASP). Senescence exacerbates cerebrovascular pathologies, but the underlying mechanisms are not understood. During EndMT, endothelial cells lose characteristic endothelial phenotypes, including some critical for cerebrovascular barrier function, and transition to a mesenchymal-like phenotype. Interestingly, EndMT can occur in response to some SASP components. This project, the subject of my dissertation work, focuses on determining if age-related increases in brain senescence drives EndMT in the brain microvasculature leading to downstream cerebrovascular fragility and cognitive decline. I have begun building skills that will be useful to this project including optimization of flow cytometry protocols, initiation of cell culture experiments, and assessment of microbleed development. I have also written a review article on EndMT and age-related diseases of the central nervous system which will be submitted for publication this month. This work aims to reveal the mechanisms underlying how senescent cells contribute to age-related cerebrovascular fragility.

- a. **Vance ML**, Gulej R, Bickel MA, Shanmugarama S, Csiszar A, Conley SM. Senescence and endothelial to mesenchymal transition in the aging cerebral vasculature. American Aging Association 52nd Annual Meeting; 2024 June; Madison, Wisconsin.
- b. **Vance ML**, Black J, Gulej R, Bickel MA, Shanmugarama S, Csiszar A, Conley SM. Senescence and endothelial to mesenchymal transition in the aging cerebral vasculature. American Aging Association 53rd Annual Meeting; 2025 May; Anchorage, Alaska.

D. Scholastic Performance

YEAR	COURSE TITLE	GRADE
UNIVERSITY OF CENTRAL OKLAHOMA - GPA 4.0		
2018	BIO 1204 Biology for Majors: Principles	A
2018	HIST 1493 History of U.S. Since 1877	A
2018	MATH 1593 Plane Trigonometry	A
2018	PSY 1103 General Psychology	A
2018	UNIV 1012 Success Central	A
2019	BIO 1225 Biology for Majors: Diversity	A
2019	BIO 1234 Environmental Biology: A Global Perspective	P*2
2019	CHEM 1103 General Chemistry I	A
2019	CHEM 1112 General Chemistry I Recitation/Lab	A
2019	MCOM 1113 Fundamentals of Speech	A
2019	BIO 2203 Cell Biology	A
2019	BIO 2211 Cell Biology Laboratory	A
2019	CHEM 1223 General Chemistry II	A
2019	CHEM 1232 General Chemistry II Recitation/Lab	A
2019	HUM 2223 General Humanities: Renaissance-Modern	A
2020	BIO 3813 Mammalian Physiology II	A
2020	CHEM 3303 Organic Chemistry I	A
2020	CHEM 3312 Organic Chemistry I Laboratory	A
2020	HLTH 1112 Healthy Life Skills	A
2020	STAT 2103 Intro Statistics for Sciences	A
2020	BIO 3803 Mammalian Physiology I	A
2020	CHEM 3323 Organic Chemistry II	A
2020	MATH 2153 Bio-Calculus	A

YEAR	COURSE TITLE	GRADE
2020	PHIL 1123 Contemporary Moral Problems	A
2021	BIO 3054 Microbiology for Majors & Lab	A
2021	BIO 4840 Capstone	P
2021	BIO 4930 Independent Study: O9-1 Cell Biology	A
2021	CHEM 3403 Biochemistry I	A
2021	GEO 2303 Regional Geography of the World	A
2021	PHY 1114 General Physics I & Lab	A
2021	BIO 3303 Genetics	A
2021	BIO 3311 Genetics Lab	A
2021	BIO 4515 Pathogenic Microbiology & Immunology & Lab	A
2021	FNRL 3483 Psychology of Grief	A
2022	AESS 3413 Creativity for Leaders	A
2022	BIO 3703 Evolution	A
2022	BIO 4134 Developmental Biology and Lab	A
2022	BIO 4930 Genetics Lab TA	A
2022	PHY 1214 General Physics II & Lab	A
UNIVERSITY OF OKLAHOMA HEALTH SCIENCES CENTER - GPA 3.955		
2022	BMSC 5031 Animal Use	S
2022	BMSC 5021 Research Methods	B
2022	BMSC 5001 Integrity Scientific Research	S
2022	BMSC 5221 Interdisciplinary First-Year Journal Club	A
2022	BMSC 6012 Molecular Systems I	A
2022	BMSC 6023 Molecular Systems II	A
2022	BMSC 6052 Cellular Systems II	A
2022	BMSC 6053 Cellular Systems I	A
2023	BIOC 6341 Signaling/Regulation (Geroscience)	A
2023	CELL 6111 Advanced Cell Biology I	A
2023	CELL 6121 Advanced Cell Biology II	A
2023	CELL 6321 Molecular and Cell Vision	A
2023	PATH 6121 Molecular Mechanisms of Human Disease	A
2023	PHSC 5581 Principles of Toxicology	A
2023	PHYO 6401 Genes to Physiology (Cellular & Physiological Aspects of Aging)	A
2023	PHYO 6401 Genes to Physiology (Angiogenesis & Lymphangiogenesis)	A
2023	PHYO 6401 Genes to Physiology (Integrative Aspects of Smooth Muscle Function)	A
2023	BMSC 5011 Experimental Design	A
2023	BMSC 6100 Bioscience Lab Rotation	A
2023	CELL 5960 Current Topics in Cell Biology	S
2023	CELL 5990 Special Studies Cell Biology	S
2023	CELL 6130 Advanced Cell Biology III	S
2024	CELL 6000 Cell Biology Lab	S
2024	CELL 5990 Special Studies Cell Biology	S
2024	CELL 5960 Current Topics in Cell Biology	S
2024	CELL 6980 Research for Drs Dissertation	X*
2024	CELL 5960 Current Topics in Cell Biology	S
2024	CELL 6980 Research for Drs Dissertation	X*

*=placeholder grade until dissertation is satisfactorily completed